CSC-185: Methods of Random Processes

Syllabus

Fall 2026

1 Course Description

The **Methods of Random Processes** course at Hofstra University introduces students to the fundamental concepts of randomness and stochastic modeling. Topics typically include probability theory, random variables, stochastic processes, and their applications in engineering, science, and data analysis. The course emphasizes both theoretical understanding and practical problem-solving skills.

This course also serves as a first step into the world of artificial intelligence, providing the probabilistic foundation needed to understand machine learning, data-driven decision making, and modern AI systems.

2 Prerequisites

Students are expected to have a solid foundation in calculus, including multivariable calculus, as well as familiarity with linear algebra. Prior exposure to basic probability concepts and programming experience is helpful but not required.

3 Logistics Details

- **Instructor:** Marco Romanelli (e-mail: marco.romanelli@hofstra.edu)
- **Office Hours**: Tuesdays 1:15 PM - 2:30 PM. More details on office hours are provided in section 8.
- **Textbook:** Though the purchase of a textbook is **not** required, students interested in further reading may consider the following references: [Ber85, BLM13, Cha23, CT06, DS12, Dev12, Gar88, GS01, Haj15, HTF09, KR18, Kle14, LR05, RW06, SSBD14, TK98, Tsy09, WMMY12, Was04, Was06].

4 Policies and Procedures

Attendance: While attendance is not mandatory, it will be recorded at the beginning of each class. Active participation is expected and will be reflected in the final grade through the evaluation of assignments. More details on the grading scheme are provided below, cf. section 4.1.

Exams: Tentative dates for exams are provided in the course schedule table, cf. section 9. Exams will cover all material presented in class up to the date of the exam. Exams after the first are non-cumulative and will focus only on the material covered since the previous exam.

4.1 Grades

Grades will be based on the following components:

- **Assignments + Participation (30%)**: There will be a series of assignments, including written problems and coding exercises, designed to reinforce the concepts covered in class. These will be graded and will contribute to (30%) of the final grade. Students are expected to submit assignments on time, and late submissions will not be accepted. Active participation in class discussions is required and will be factored into this portion of the grade.
- **Tests (30%)**: There will be two midterm exams. The lowest test grade will be dropped, and the remaining will be averaged to contribute to the final grade.
- **Final Exam (40%)**: The final exam will be comprehensive, covering all material presented throughout the course. It will test students' understanding of key concepts, their ability to apply these concepts to solve problems, and their proficiency in interpreting results.

Exam Conflicts: There are no makeup or redo opportunities; however, the lowest exam grade will be dropped.

5 Classroom Conduct

Respectful Behavior and Professionalism are expected at all times. Students are expected to take notes and engage during class.

Huge efforts have been made to prepare this course, and all the materials and resources provided in class and shared online are sufficient to pass the exams. Recording, photographing, or any other unauthorized use of electronic devices resulting in a disruption of the class will be considered a violation of the classroom conduct policy and will result in disciplinary action.

6 Academic Integrity

Plagiarism and other forms of academic dishonesty are serious offenses.

Similarly, late online submissions will not be accepted. During in-person exams, after announcing that the time is up, students have one minute to stop writing and submit their work. Any work submitted after the one-minute grace period will be collected separately and will be penalized by a deduction of 15 points from the final grade for each occurrence. This policy is in place to ensure fairness and consistency for all students.

Students are expected to adhere to the highest ethical and professional standards. For detailed policies on academic integrity, including violations, procedures, and appeal rights, refer to **Faculty Policy Series #11** (undergraduate students) and **Faculty Policy Series #11G** (graduate students).

7 Course Materials

Course Resources: Shared materials include lecture notes in the form of slides and practice problems.

In addition, autograded coding assignments will be provided to help students practice and apply the concepts covered in class. These resources are intended to reinforce understanding and support exam preparation.

Written assignments will be graded online, and solutions will be reviewed in class so that students can ask questions and clarify any uncertainties about the material.

These materials are sufficient for strong performance on the exams. Students who are interested in additional resources, are invited to consult the suggested textbooks, materials available at the library, and online.

8 Office Hours

Office hours are held on Tuesdays from 1:20 PM to 2:30 PM. These sessions are intended to address specific, well-defined questions that cannot be resolved through other means. To ensure that time is used effectively, meetings are limited to 15 minutes per student and are scheduled on a first-come, first-served basis. A sign-up sheet will be made available on my office door (SIC 215), and students are encouraged to sign up in advance to secure a time slot. Walk-ins may be accommodated if time permits, but priority will be given to those who have signed up.

Please register only if you are certain you can attend. Students must complete the form available in the "Resources for Students" section on Canvas, clearly indicating the concepts or materials they need help with. This allows for more focused and productive meetings.

Because office hours are in high demand and time is limited, missed appointments affect other students' ability to receive support. For this reason, repeated absences or cancellations (2 or more) without adequate notice will result in a deduction of 5 points from the final grade for each occurrence. If you are unable to attend a scheduled session, please provide at least 24 hours' notice so the time can be offered to another student.

9 Tentative Schedule

Date	Topic	References	Assignments/Notes
Tue, Sep 1, 2026	L00 - First Day of Class: – Intro to the course – Syllabus Overview		GradeScope Init
Thu, Sep 3, 2026	L01 - Intro to Statistical Inference: – Statistical Inference – Samples & Populations – Role of Probability	[WMMY12, §1.1-1.6]	Quiz #01 Release
Tue, Sep 8, 2026	L02 - Probability: – Add./Mul. Rules – Counting Events – Sample Spaces	[Dev12, §2.1-2.3]	

Date	Topic	References	Assignments/Notes
Thu, Sep 10, 2026	L03 - Probability: – Conditional Probability – Independence	[Dev12, §2.4]	Quiz #03 Release Code #03 Release
Tue, Sep 15, 2026	L04 - Random Variables: – Bayes' Theorem – Random Variables – Discrete Prob. Distributions	[Dev12, §2.4] [WMMY12, §2.7] [HTF09, §6.6.3]	
Thu, Sep 17, 2026	L05 - Measures of information: – Entropy – Mutual Information – Divergences	[CT06, §2.1 - 2.10]	Code #04 Release
Tue, Sep 22, 2026	L06 - Continuous Probability: – Continuous Prob. Distributions – Joint Prob. Distributions	[Dev12, §4.1 - 4.3] [WMMY12, §3.3 - 3.4]	
Thu, Sep 24, 2026	L07 - Mathematical Expectation: – Mean of a Random Variable – Variance of a Random Variable – Chebyshev's Theorem	[Kle14, §5.2] [WMMY12, §4.1 - 4.4]	Quiz #05 Release Code #05 Release
Tue, Sep 29, 2026	L08 - The Normal Distribution: – Formal Definition – Area under the curve – Properties	[WMMY12, §6.1-6.3]	Quiz #06 Release Quiz #07 Release
Thu, Oct 1, 2026	L09 - Sampling Distributions: – Sampling Distributions – Central Limit Theorem	[Kle14, §15.5] [LR05, §11.2]	
Tue, Oct 6, 2026			Mid-Semester break
Thu, Oct 8, 2026			Test #1
Tue, Oct 13, 2026	L10 - Estimation from Samples: – Confidence Intervals – Uncertainty Quantification	[WMMY12, §9.1 -9.4]	
Thu, Oct 15, 2026	L11 - Estimation from Samples: – Maximum Likelihood Estimation	[HTF09, §8.2] [LR05, §1.7]	Quiz #10 Release
Tue, Oct 20, 2026	L12 - Hypothesis Testing: – Null and Alternative Hypotheses – Type I and Type II Errors	[WMMY12, §10.1 - 10.2]	Mid-Semester Advisories
Thu, Oct 22, 2026	L13 - Hypothesis Testing: – Use of P-values – Choice of Sample Size	[LR05, §3.1 - 3.3]	Quiz #11 Release Code #11 Release
Tue, Oct 27, 2026	L14 - Hypothesis Testing in AI: – Mahalanobis Distance – Testing distributional shift	[HTF09, §12.5]	
Thu, Oct 29, 2026	L15 - Linear Regression (pt. 1)	[WMMY12, §11.1 - 11.3]	Code #12 Release
Tue, Nov 3, 2026	L16 - Linear Regression (pt. 2)	[HTF09, §3.1-3.2]	
Thu, Nov 5, 2026	*L17 - Learning (pt. 1) – The Learning Problem – Theoretical Bounds against Overfitting	[SSBD14, §2.1 - 2.3]	Code #14 Release Quiz #16 Release Code #16 Release
Tue, Nov 10, 2026	*L18 - Learning (pt. 2) – Empirical Risk Minimization – PAC Learning	[SSBD14, §2.4]	
Thu, Nov 12, 2026			Test #2
Tue, Nov 17, 2026	*L19 - Intro to ML: ANN and SGD (pt. 1) – Artificial Neural Networks – Stochastic Gradient Descent – Application of the ERM	[SSBD14, §14.3]	
Thu, Nov 19, 2026	*L20 - Intro to ML: ANN and SGD (pt. 2) – Chain Rule for Backpropagation – Parameter Fitting	[SSBD14, §20.6]	
Tue, Nov 24, 2026	*L21 - Gaussian Processes – Introduction to Gaussian Processes – Regression with Gaussian Processes – Classification with Gaussian Processes	[RW06, §2.1-3.4]	
Thu, Nov 26, 2026			Thanksgiving Break

Date	Topic	References	Assignments/Notes
Tue, Dec 1, 2026	*L22 - Concentration Inequalities – Hoeffding's Inequality	[BLM13, §2.6-2.7]	
Thu, Dec 3, 2026	*L23 - Concentration Inequalities – Implications of Hoeffding's Inequality – Bernstein's Inequality	[BLM13, §2.7-2.8]	
Tue, Dec 8, 2026	*L24 - Bayesian Reasoning – Why Data Matters – Bayesian Prediction	[SSBD14, §24.5]	

*Topics marked with a star are reserved for the final part of the course and may be skipped if time does not allow for their coverage. They are more advanced and theoretical. However, students who are interested in these topics and would like to get a head start to understand basic concepts in machine learning and modern AI systems are encouraged to review the materials provided in class and consult the suggested references.

References

[Ber85] James O. Berger. *Statistical Decision Theory and Bayesian Analysis*. Springer Series in Statistics. Springer, 1985.

[BLM13] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.

[Cha23] Stanley Chan. *Introduction to Probability for Data Science*. Michigan Publishing Services, 2023.

[CT06] Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2nd edition, 2006.

[Dev12] Jay L. Devore. *Probability and Statistics for Engineering and the Sciences*. Cengage Learning, 2012.

[DS12] Morris H. DeGroot and Mark J. Schervish. *Probability and Statistics*. Pearson, 2012.

[Gar88] William A. Gardner. Introduction to random processes with applications to signals and systems. *Automatica*, 24(5):721–722, 09 1988.

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[Haj15] Bruce Hajek. *Random Processes for Engineers*. Cambridge University Press, 2015.

[HTF09] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2009.

[Kle14] Achim Klenke. *Probability Theory: A Comprehensive Course*. Universitext. Springer London, 2014.

[KR18] Rajeeva L. Karandikar and B. V. Rao. *Introduction to Stochastic Calculus*. Indian Statistical Institute Series. Springer Singapore, 2018.

[LR05] Erich Leo Lehmann and Joseph P Romano. *Testing statistical hypotheses*. Springer Science & Business Media, 2005.

[RW06] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian processes for machine learning*. Adaptive computation and machine learning. MIT Press, 2006.

[SSBD14] Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning - From Theory to Algorithms*. Cambridge University Press, 2014.

[TK98] Howard M Taylor and Samuel Karlin. *An Introduction to Stochastic Modeling*. Academic Press, San Diego, 1998.

[Tsy09] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, 2009.

[Was04] Larry Wasserman. *All of Statistics: A Concise Course in Statistical Inference*. Springer Texts in Statistics. Springer, 2004.

[Was06] Larry Wasserman. *All of Nonparametric Statistics*. Springer Texts in Statistics. Springer, 2006.

[WMMY12] Ronald E Walpole, Raymond H Myers, Sharon L Myers, and Keying Ye. *Probability & Statistics for Engineers & Scientists*. Pearson, Boston, 9th edition, 2012.